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(54) **PARK ASSIST SYSTEM**

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(57) **ABSTRACT**

A park assist system stores, as a registered travel path, a travel path of a vehicle when causing the vehicle to enter a predetermined parking space or a travel path of the vehicle when causing the vehicle to exit the parking space, and performs automatic park control in which the vehicle is caused to travel autonomously based on the registered travel path, when causing the vehicle to enter or exit the parking space from next time onwards. The park assist system includes a determination unit that determines whether to store the registered travel path as an entry path or an exit path, based on a state of the vehicle when a registration request to store the registered travel path is made, and determines whether to store a travel path of the vehicle immediately before or a travel path of the vehicle immediately after when the registration request is made.

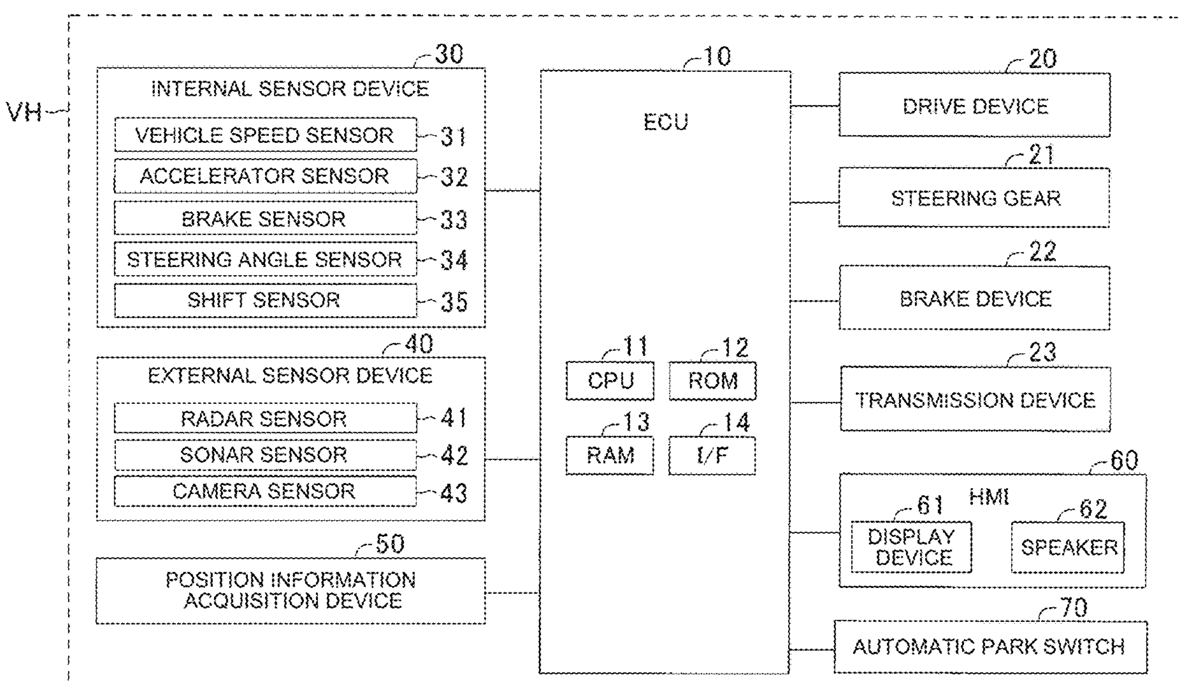
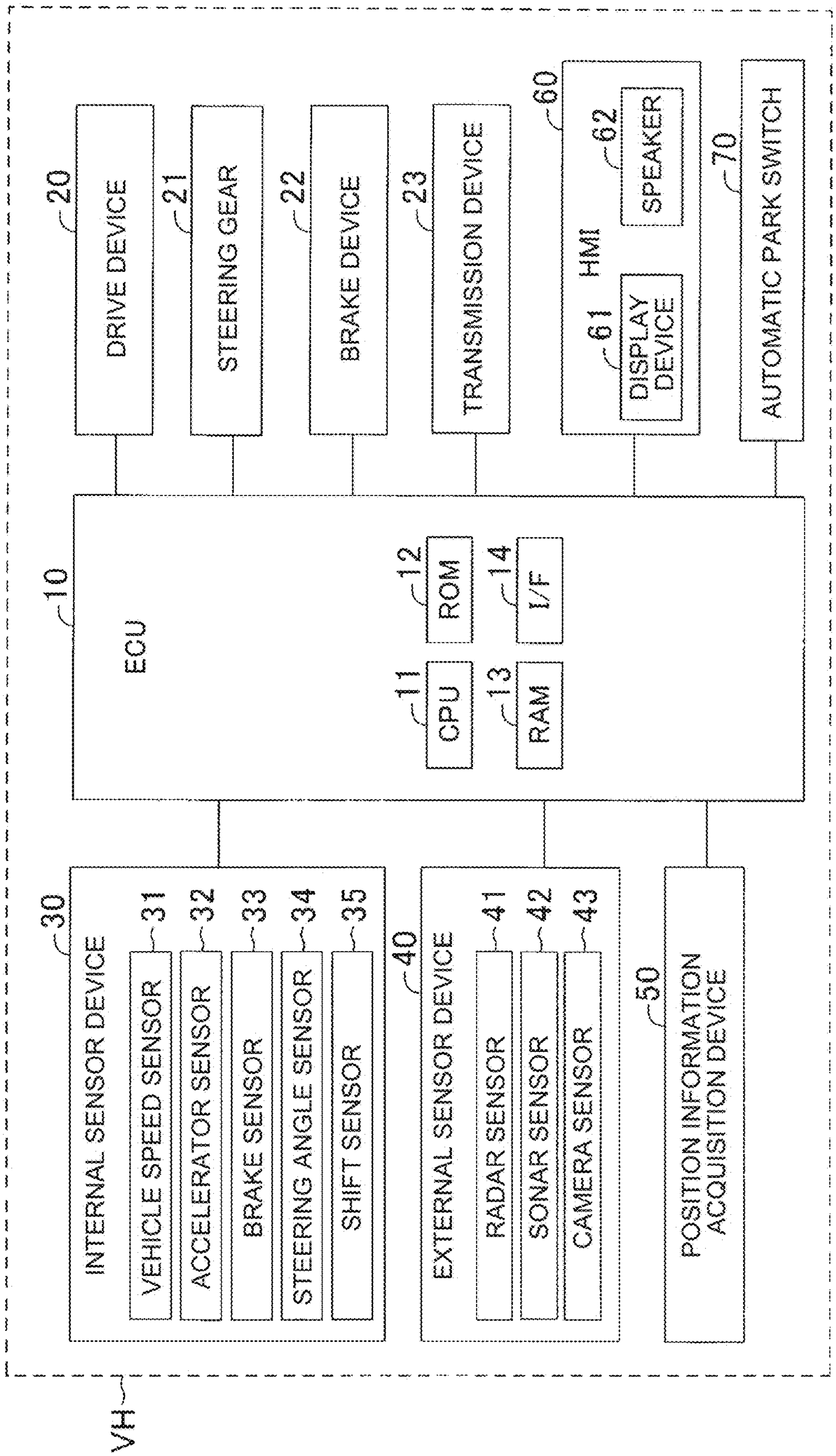


FIG. 1



VH
20 DRIVE DEVICE
21 STEERING GEAR
22 BRAKE DEVICE
23 TRANSMISSION DEVICE
60 HMI
61 DISPLAY DEVICE
62 SPEAKER
70 AUTOMATIC PARK SWITCH
10 ECU
11 CPU
12 ROM
13 RAM
14 I/F
30 INTERNAL SENSOR DEVICE
31 VEHICLE SPEED SENSOR
32 ACCELERATOR SENSOR
33 BRAKE SENSOR
34 STEERING ANGLE SENSOR
35 SHIFT SENSOR
40 EXTERNAL SENSOR DEVICE
41 RADAR SENSOR
42 SONAR SENSOR
43 CAMERA SENSOR
50 POSITION INFORMATION ACQUISITION DEVICE

FIG. 2

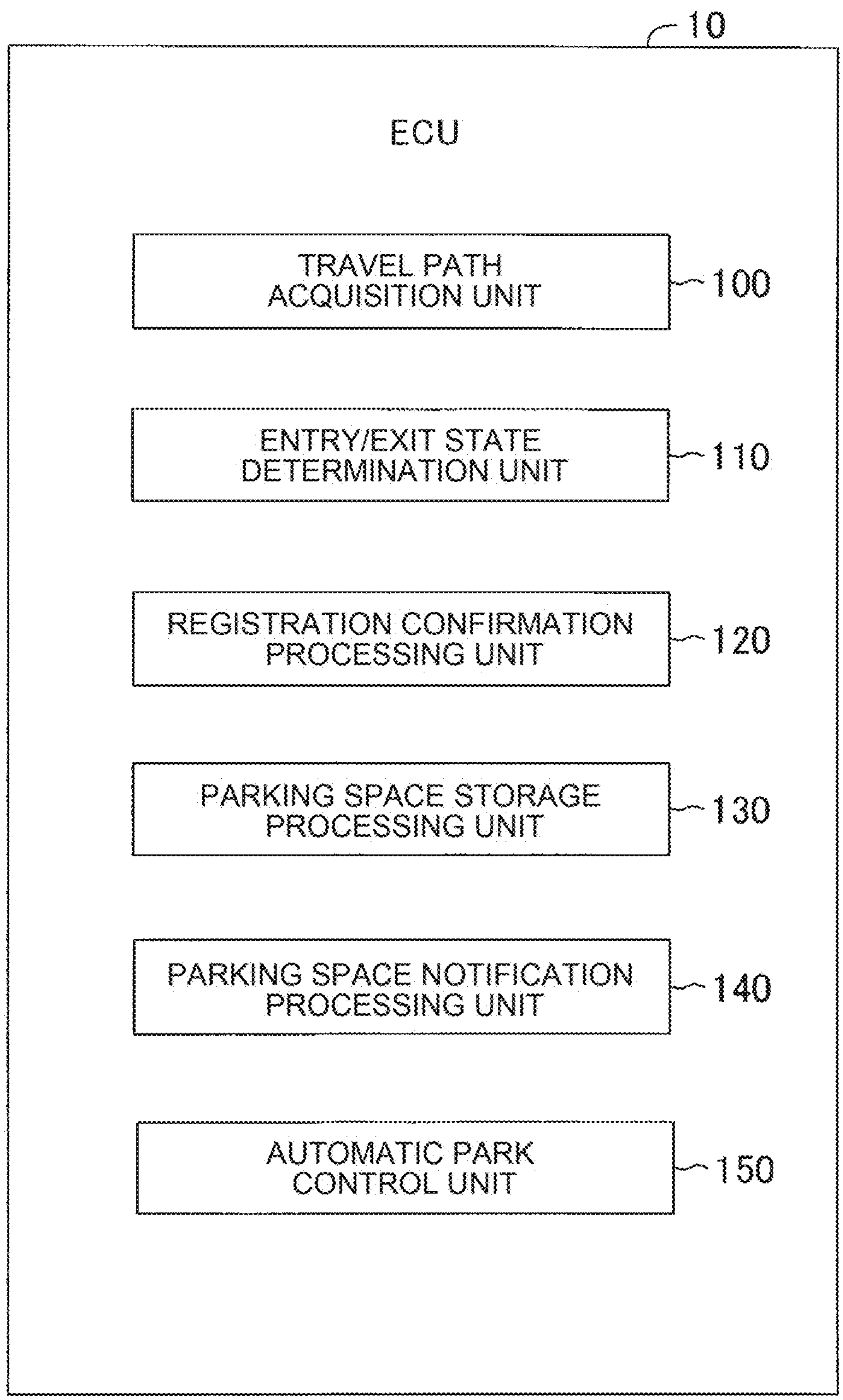

10
ECU
TRAVEL PATH ACQUISITION UNIT
100
ENTRY/EXIT STATE DETERMINATION UNIT
110
REGISTRATION CONFIRMATION PROCESSING UNIT
120
PARKING SPACE STORAGE PROCESSING UNIT
130
PARKING SPACE NOTIFICATION PROCESSING UNIT
140
AUTOMATIC PARK CONTROL UNIT
150

FIG. 3

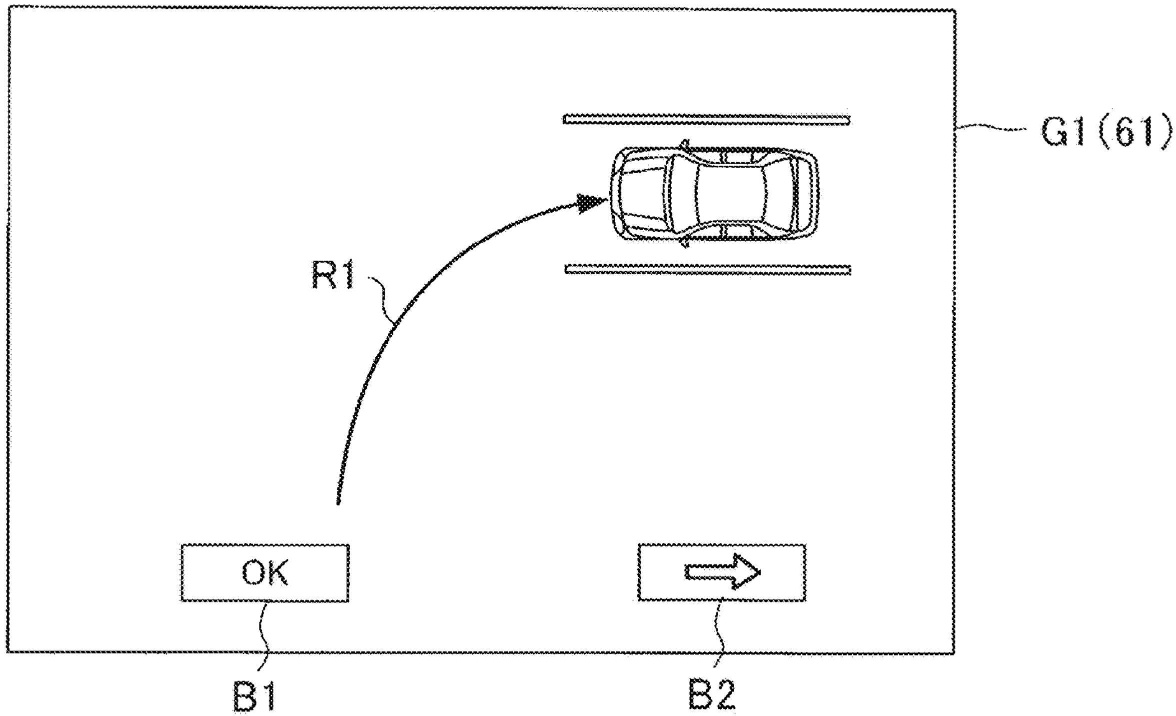


FIG. 4

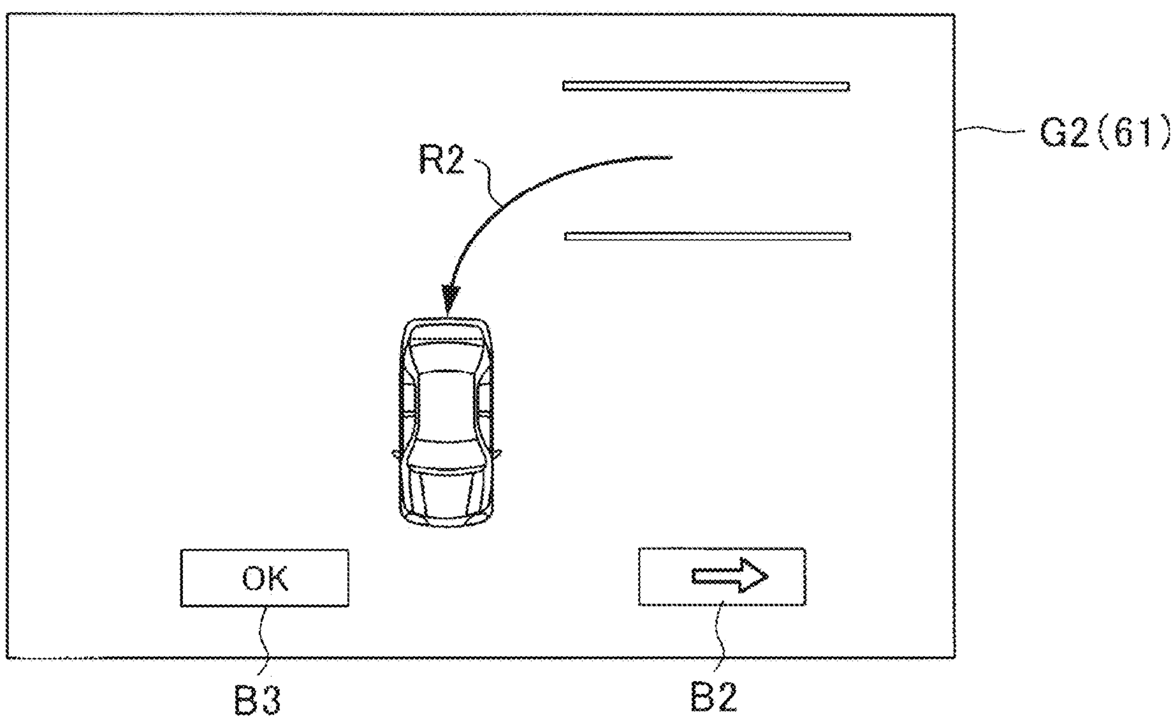


FIG. 5A

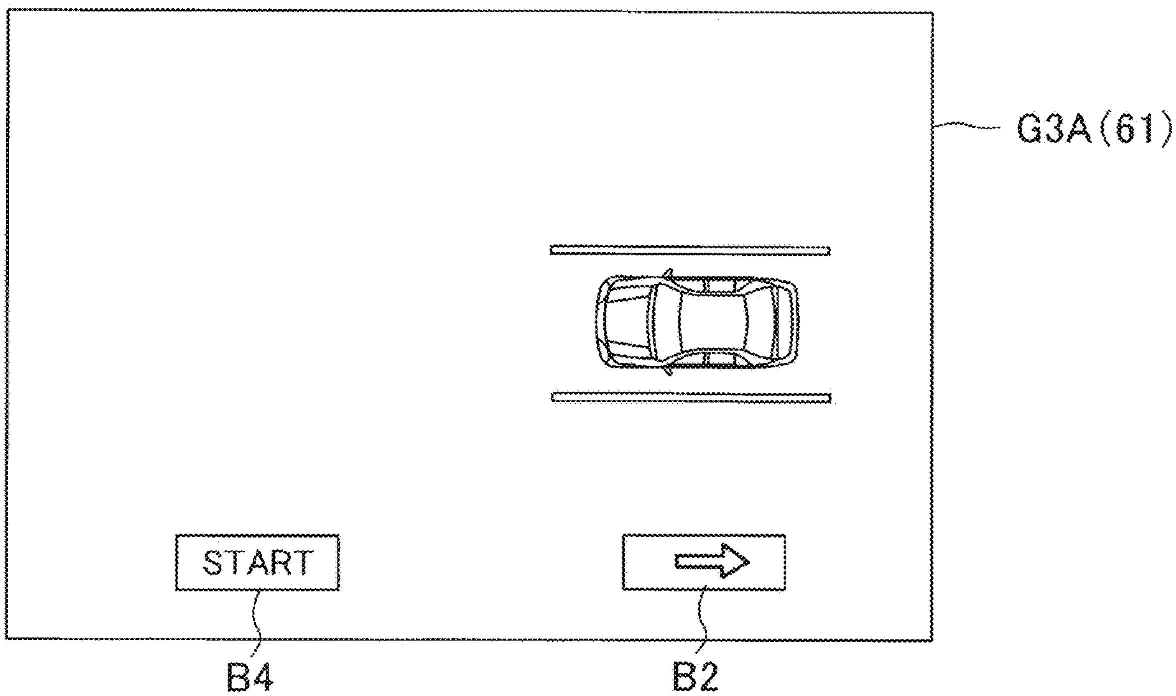


FIG. 5B

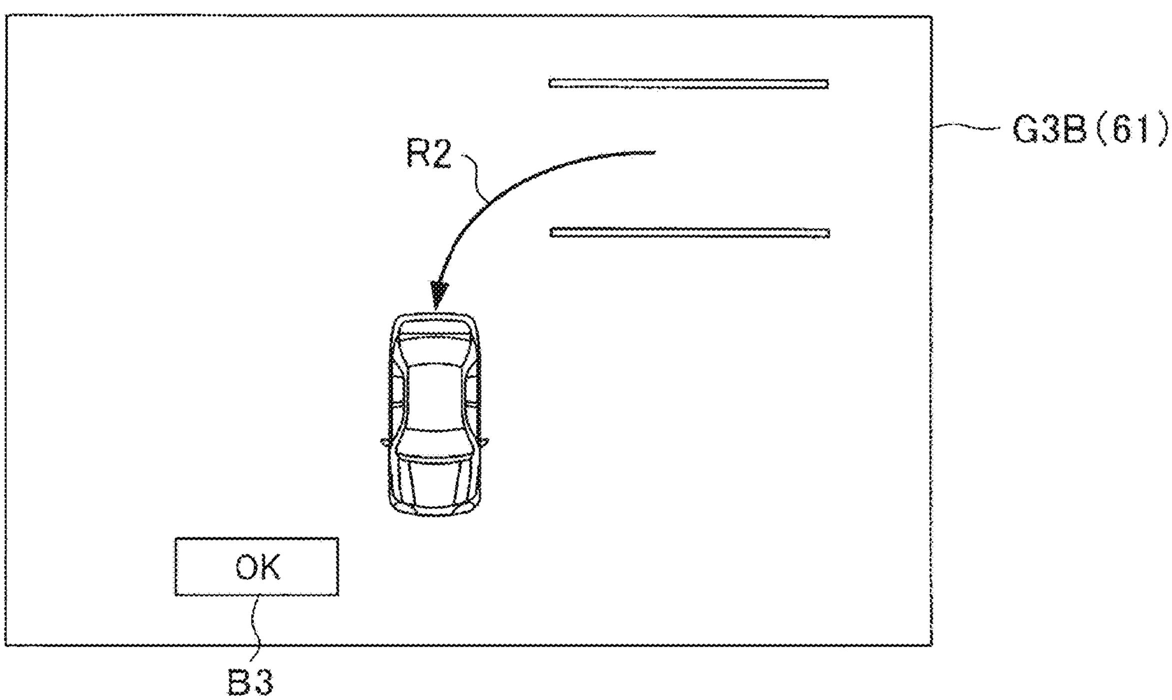


FIG. 6A

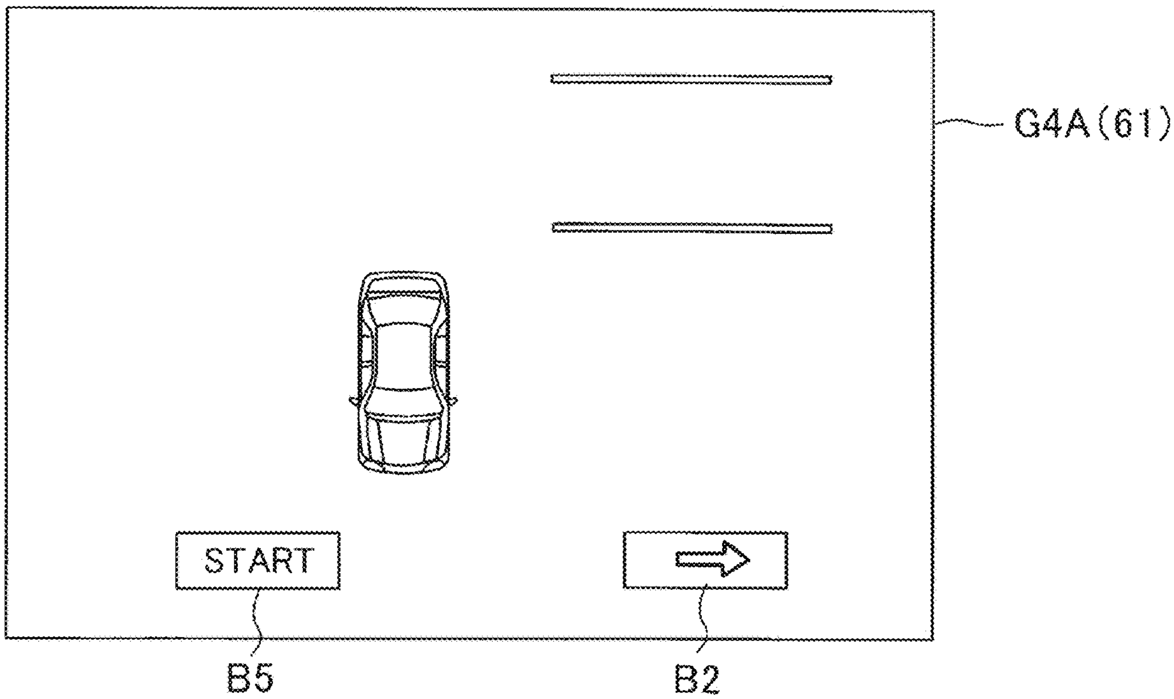


FIG. 6B

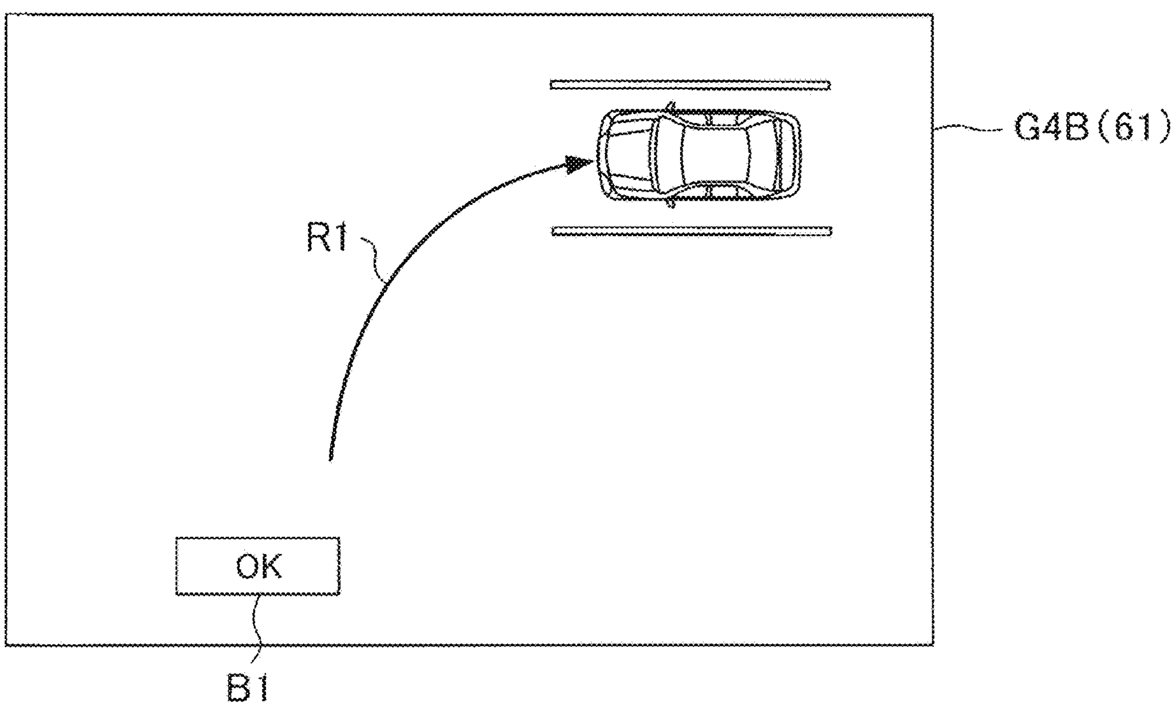
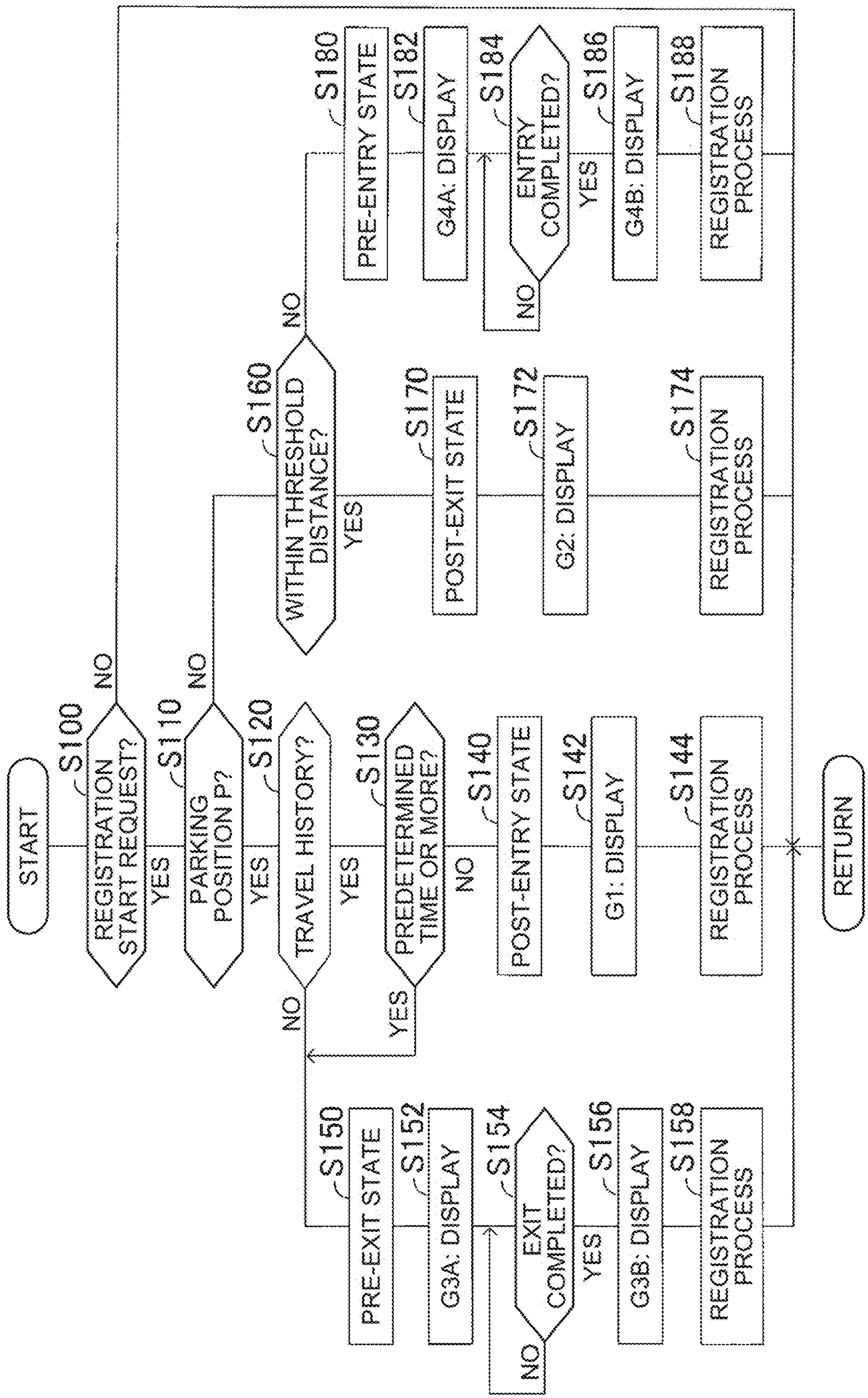


FIG. 7


START
S100
REGISTRATION START REQUEST?
YES
NO
S110
PARKING POSITION P?
YES
NO
S120
TRAVEL HISTORY?
YES
NO
S130
PREDETERMINED TIME OR MORE?
YES
NO
S140
POST-ENTRY STATE
S142
G1: DISPLAY
S144
REGISTRATION PROCESS
S150
PRE-EXIT STATE
S152
G3A: DISPLAY
S154
EXIT COMPLETED?
YES
NO
S156
G3B: DISPLAY
S158
REGISTRATION PROCESS
S160
WITHIN THRESHOLD DISTANCE?
YES
NO
S170
POST-EXIT STATE
S172
G2: DISPLAY
S174
REGISTRATION PROCESS
S180
PRE-ENTRY STATE
S182
G4A: DISPLAY
S184
ENTRY COMPLETED?
YES
NO
S186
G4B: DISPLAY
S188
REGISTRATION PROCESS
RETURN

PARK ASSIST SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-026695 filed on Feb. 26, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to park assist systems.

2. Description of Related Art

[0003] For example, Japanese Unexamined Patent Application Publication No. 2023-125795 (JP 2023-125795 A) discloses a park assist system that stores, as a target parking space for park assistance, position information of a parking space where a vehicle is frequently parked. The park assist system disclosed in JP 2023-125795 A performs automatic control of park assistance for parking the vehicle in the target parking space, when it detects that the vehicle has approached the target parking space.

SUMMARY

[0004] In recent years, path memory park assist systems have been used in practical applications. A path memory park assist system not only registers position information of a parking space but also stores a travel path a vehicle traveled when a driver parked in this parking space. When the vehicle is to be parked in the same parking space from next time onwards, the path memory park assist system causes the vehicle to autonomously drive along the stored travel path. When such a park assist system stores a travel path, it is necessary to identify whether this travel path is a path to enter the parking space or a path to exit the parking space, or whether the travel path to be registered is a travel path the vehicle has traveled immediately before or a travel path that the vehicle is going to travel. Performing these identifications by driver's selection operations is disadvantageous because such operations are complicated enough to bother the driver. That is, it can be said that there is room for improvement in convenience.

[0005] The present disclosure was made to solve the above issue, and an object of the present disclosure is to improve convenience by simplifying an operation in path memory park assistance.

[0006] A park assist system of the present disclosure is configured to store, as a registered travel path, a travel path of a vehicle when causing the vehicle to enter a predetermined parking space or a travel path of the vehicle when causing the vehicle to exit the parking space, and perform automatic park control when causing the vehicle to enter or exit the parking space from next time onwards. The automatic park control is control in which the vehicle is caused to travel autonomously based on the registered travel path. The park assist system includes a determination unit configured to

- **[0007]** determine whether to store the registered travel path as an entry path or an exit path, based on a state of the vehicle when a registration request to store the registered travel path is made, and
- **[0008]** determine whether to store a travel path of the vehicle immediately before or a travel path of the vehicle immediately after when the registration request is made.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0010] FIG. **1** is a schematic diagram illustrating a hardware configuration of a vehicle according to an embodiment;

[0011] FIG. **2** is a schematic diagram illustrating a software configuration of a control device according to the embodiment;

[0012] FIG. **3** is a schematic diagram illustrating an example of a registration confirmation screen according to the embodiment;

[0013] FIG. **4** is a schematic diagram illustrating an example of a registration confirmation screen according to the embodiment;

[0014] FIG. **5**A is a schematic diagram illustrating a registration confirmation screen according to the embodiment;

[0015] FIG. **5**B is a schematic diagram illustrating a registration confirmation screen according to the embodiment;

[0016] FIG. **6**A is a schematic diagram illustrating a registration confirmation screen according to the embodiment;

[0017] FIG. **6**B is a schematic diagram illustrating a registration confirmation screen according to the embodiment; and

[0018] FIG. **7** is a flowchart for explaining routines of the entry/exit state determination process and the registration confirmation process according to the present embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0019] Hereinafter, a parking assist system according to the present embodiment will be described with reference to the drawings.

Hardware Configuration

[0020] FIG. **1** is a schematic diagram illustrating a hardware configuration of a vehicle VH according to the present embodiment. The vehicle VH has ECU (Electronic Control Unit) **10**. ECU **10** includes CPU (Central Processing Unit) **11**, ROM (Read Only Memory) **12**, RAM (Random Access Memory) **13**, and interface device **14**. CPU **11** is a processor that executes various programs stored in ROM **12**. ROM **12** is a non-volatile memory that stores data and the like required for CPU **11** to execute various programs. RAM **13** provides a working area to be deployed when various programs are executed by CPU **11**. The interface device **14** is a communication device for communicating with an external device.

[0021] ECU **10** is a central device that performs automatic entry control for automatically causing the vehicle VH to enter the parking space and automatic exit control for automatically causing the vehicle VH to exit the parking space (hereinafter, these automatic entry control and automatic exit control may be collectively referred to as "automatic park control"). In the present embodiment, the automatic park control is performed on a parking space registered in advance by the driver, such as a parking space having a relatively high frequency of use. Hereinafter, the parking space that has been registered will be referred to as "registered parking space". The target parking space can be registered as a registered parking space as long as the parking space is not only a parking space whose parking frame is defined by a white line etc, but any parking space in which a space in which a vehicle VH can enter can be secured. The registered parking space is registered by the driver manually acquiring a travel path when the vehicle VH is caused to enter or exit the parking space and storing the acquired travel path in association with the position information of the parking space.

[0022] The automatic entry control of the vehicle VH to the registered parking space and the automatic exit control from the registered parking space are performed by causing the vehicle VH to automatically travel along the travel path stored in advance. For this reason, the drive device **20**, the steering device **21**, the braking device **22**, the transmission device **23**, the internal sensor device **30**, the external sensor device **40**, the position information acquisition device **50**, HMI (Human Machine Interface) **60**, the automatic park switch **70**, and the like are communicably connected to ECU **10**.

[0023] The drive device **20** generates a driving force to be transmitted to the driving wheels of the vehicle VH. Examples of the drive device **20** include an electric motor and an engine. In the present embodiment, the vehicle VH may be any of a hybrid electric vehicle (HEV), a plug-in Hybrid vehicle (PHEV), a fuel cell electric vehicle (FCEV), a battery electric vehicle (BEV), and an engine-driven vehicle. The steering device **21** applies a steering force to the wheels of the vehicle VH. The braking device **22** applies a braking force to the wheels of the vehicle VH. The transmission device **23** is configured to be selectively switchable to a parking range that locks the rotation of the drive wheels, a reverse range that causes the vehicle VH to travel backward, a neutral range that blocks the transmission of power, and a drive range that causes the vehicle VH to travel forward.

[0024] The internal sensor device **30** is a sensor for detecting the condition of VH. Specifically, the internal sensor device **30** includes a vehicle speed sensor **31**, an accelerator sensor **32**, a brake sensor **33**, a steering angle sensor **34**, a shift sensor **35**, and the like. The vehicle speed sensor **31** detects the traveling speed of the vehicle VH, that is, the vehicle speed. The accelerator sensor **32** detects an operation amount of an accelerator pedal (not shown) by a driver. The brake sensor **33** detects an operation amount of a brake pedal (not shown) by the driver. The steering angle sensor **34** detects a rotation angle of a steering wheel or a steering shaft (not shown), that is, a steering angle. The shift sensor **35** detects a shift position (parking position P, reverse position R, neutral position N, drive position D, and the like) of the transmission device **23**. The internal sensor device **30** transmits the condition of the vehicle VH detected by the sensors **31** to **35** to ECU **10** at a predetermined cycle.

[0025] The external sensor device **40** is a sensor or the like that recognizes target information related to a target in the vicinity of VH. Specifically, the external sensor device **40** includes a radar sensor **41**, a sonar sensor **42**, a camera sensor **43**, and the like. The radar sensor **41** includes a millimeter wave radar and/or a lidar. The millimeter-wave radar radiates a radio wave in a millimeter-wave band, and receives a reflected wave reflected by a target existing in the radiation range, thereby obtaining a relative distance, a relative velocity, and the like between the vehicle VH and the target. The lidar sequentially scans the pulsed laser beam having a wavelength shorter than the millimeter wave toward a plurality of directions, and receives the reflected light reflected by the target, thereby obtaining a shape of the target, a relative distance between the vehicle VH and the target, a relative velocity, and the like. The sonar sensor **42** emits an ultrasonic wave and receives a reflected wave reflected by a target existing in the emission area, thereby obtaining a relative distance, a relative velocity, and the like between the vehicle VH and the target. The camera sensor **43** is, for example, a stereo camera or a monocular camera, and a digital camera having an image sensor such as a CMOS or a CCD can be used. The camera sensor **43** processes the captured image data to acquire the shape of the target object, the relative distance between the vehicle VH and the target object, the relative speed, and the like. The external sensor device **40** transmits the acquired target object data to ECU **10** at a predetermined cycle. Note that the external sensor device **40** does not necessarily have to include all of the radar sensor **41**, the sonar sensor **42**, and the camera sensor **43**, and may include, for example, only the camera sensor **43**.

[0026] The position information acquisition device **50** acquires the present position information of the vehicle VH. As the position-information acquisition device **50**, for example, a GPS (Global Positioning System), GNSS (Global Navigation Satellite System) or the like included in a navigation device (not shown) can be used. The position information acquisition device **50** sends the acquired present position information of the vehicle VH to **30** ECU **10** at a predetermined cycle.

[0027] HMI **60** is an interface for inputting and outputting data between ECU **10** and drivers, and includes an input device and an output device. Examples of the input device include a touch panel type liquid crystal display, a switch, and a sound pickup microphone. Examples of the output device include a display device **61** and a speaker **62**. The display device **61** is, for example, a center display, a multi-information display, or the like. The speaker **62** is, for example, a speaker of an acoustic system or a navigation system.

[0028] The automatic park switch **70** is, for example, a switch provided in a center console, an instrument panel, or the like of the vehicle VH and ON operated by an occupant (for example, a driver) of the vehicle VH. When the automatic park switch **70** is turned ON, ECU **10** receives the automatic park start request. Note that the automatic park switch **70** is not limited to the physical switch, and may be a touch-type switch image displayed on the display device **61**. In addition, the automatic park start request may be acquired by voice recognition using a voice collection microphone or may be acquired by gesture recognition using a driver camera.

Software Configuration

[0029] FIG. **2** is a schematic diagram illustrating a software configuration of ECU **10** according to the present embodiment. As illustrated in FIG. **2**, ECU **10** includes, as functional elements, a travel path acquisition unit **100**, an entry/exit state determination unit **110**, a registration confirmation processing unit **120**, a parking space storage processing unit **130**, a parking space notification processing unit **140**, an automatic park control unit **150**, and the like. These functional elements **100** to **150** are realized by CPU **11** of ECU **10** reading a program stored in ROM **12** into a RAM **13** and executing the program. Note that all or a part of the functional elements **100** to **150** may be provided in another ECU separate from ECU **10** or in an information processing device of a facility (e.g., a control center) capable of communicating with the vehicle VH.

[0030] The travel path acquisition unit **100** acquires a travel path of the vehicle VH at the nearest predetermined distance, and temporarily stores the travel path in, for example, a RAM **13**. The travel path acquisition unit **100** sequentially deletes the travel paths acquired prior to the predetermined distance and stored in RAM **13** or the like as the new travel path is acquired. When the power switch (or the ignition switch) of the vehicle VH is turned OFF, the travel path acquisition unit **100** deletes (resets) all the travel paths stored in RAM **13** or the like. The predetermined distance is not particularly limited, but may be set based on a distance (e.g., several tens of meters) required for the vehicle VH to enter or exit the predetermined parking space.

[0031] Here, the travel path of the vehicle VH refers to a moving trajectory of the vehicle position that changes as the vehicle VH travels. The vehicle position is, for example, a predetermined geometric center position in a plan view of the vehicle VH. The vehicle position may be another position of the vehicle VH (for example, a center position of the left and right front wheels or a center position of the left and right rear wheels). The moving trajectory of the vehicle VH may be acquired, for example, by odometry based on the detection results of the vehicle speed sensor **31** and the steering angle sensor **34**. Alternatively, the moving trajectory of the vehicle VH may be used in combination with the position information of the vehicle VH acquired by the position information acquisition device **50**. Alternatively, the moving trajectory of the vehicle VH may be acquired by optical flow based on road surface images or the like of the surroundings of the vehicle VH captured by the camera sensor **43**.

[0032] Upon receiving the registration start request from the driver, the entry/exit state determination unit **110** performs the entry/exit state determination process. The entry/exit state determination process determines whether the vehicle VH is in the "pre-entry state" immediately before the vehicle VH enters the predetermined parking space, or is in the "pre-exit state" immediately before the vehicle VH exits the predetermined parking space, or is in the "post-entry state" immediately after the vehicle VH enters the predetermined parking space, or is in the "post-exit state" immediately after the vehicle VH exits the predetermined parking space. The driver registration start request may be acquired based on, for example, a touch operation of a switch image displayed on the display device **61**. Alternatively, it is only necessary to acquire the registration start request from the driver based on an operation of turning on a physical switch provided in a center console, an instrument panel, or the like. Alternatively, the registration start request from the driver may be acquired on the basis of voice recognition by the voice pickup microphone or the like. Details of the entry/exit state determination processing performed by the entry/exit state determination unit **110** will be described below.

[0033] When the registration start request from the driver is received, the entry/exit state determination unit **110** determines that the vehicle VH is in the "post-entry state" when the first specific condition is satisfied. The first specific condition is that the shift position is the parking position P and there is a travel history (a travel path acquired by the travel path acquisition unit **100**) immediately before. When receiving the registration start request from the driver, if the second specific condition is satisfied, the entry/exit state determination unit **110** determines that the vehicle VH is in the "pre-exit state". The second specific condition is that the shift position is the parking position P and there is no travel history (travel path acquired by the travel path acquisition unit **100**) immediately before. When the third specific condition is satisfied upon receipt of the registration-start request from the driver, the entry/exit state determination unit **110** determines that the vehicle VH is in the "post-exit state". The third specific condition is that the shift position is a position other than the parking position P (for example, the drive position D), and the travel distance of the vehicle VH after the shift position is switched from the parking position P is less than or equal to the predetermined threshold distance. When the fourth specific condition is satisfied upon receipt of the registration start request from the driver, the entry/exit state determination unit **110** determines that the vehicle VH is in the "pre-entry state". The fourth specific condition is that the condition of the vehicle VH does not correspond to any of the first specific condition, the second specific condition, and the third specific condition. When the fourth specific condition is satisfied, specifically, the shift position is a position other than the parking position P, and the travel distance of the vehicle VH after the shift position is switched from the parking position P exceeds a predetermined threshold distance.

[0034] Here, a case where a registration start request is made after the driver switches the shift position to the parking position P will be considered. In a case where the period from the switching to the parking position P to the registration start request is short (for example, several seconds), it is estimated that the driver desires to register the immediately preceding travel path as the entry path. However, when the period from the switch to the parking position P to the registration start request is relatively long (for example, several tens of seconds, several minutes), it is estimated that the driver does not register the entry path, but wants to register the exit path for driving the vehicle VH. Such scenes include, for example, a case where the driver restarts after stopping the vehicle VH on the road shoulder, a case where the driver parks the vehicle VH in a parking lot such as a shop, and then restarts without turning off the power switch. Even when the shift position is the parking position P and there is a travel history (travel path acquired by the travel path acquisition unit **100**) immediately before, the entry/exit state determination unit **110** determines that the vehicle VH is in the "pre-exit state" when the fifth specific condition is satisfied. The fifth specific condition is that a predetermined time or more has elapsed from the time when the shift position is switched to the parking position P until the reception of the registration start request of the driver.

[0035] The registration confirmation processing unit **120** performs registration confirmation processing for displaying a registration confirmation screen of the registration mode corresponding to the determination result from the entry/exit state determination unit **110** on the display device **61**. Details of the registration confirmation processing performed by the registration confirmation processing unit **120** will be described below.

[0036] The registration confirmation processing unit **120** displays the post-entry registration confirmation screen G**1** shown in FIG. **3** on the display device **61** when the entry/exit state determination unit **110** determines that the post-entry state is satisfied (that is, when the first specific condition is satisfied). The post-entry registration confirmation screen G**1** includes, for example, an entry path R**1** schematically showing a part required for entry of the latest travel path stored in RAM **13** or the like by the travel path acquisition unit **100**, an entry registration confirmation button B**1**, and a switching button B**2**. When the driver touches the entry registration confirmation button B**1**, the registration confirmation processing unit **120** transmits a registration command for storing the entry path R**1** in association with the position information of the parking space to the parking space storage processing unit **130**. On the other hand, when the driver touches the switching button B**2**, the registration confirmation processing unit **120** displays a registration confirmation screen of another registration mode on the display device **61**. The registration confirmation screen of another registration mode is, for example, a post-exit registration confirmation screen G**2**, a pre-exit registration confirmation screen G**3**A, and a pre-entry registration confirmation screen G**4**A, which will be described later.

[0037] As described above, when the entry/exit state determination unit **110** determines the post-entry state, a post-entry registration confirmation screen G**1** including the post-entry registration confirmation button B**1** and the entry path R**1** is displayed on the display device **61**. As a result, the driver can easily perform the entry registration without selecting whether the registration operation is before or after entry, or whether the travel path is the entry path or the exit path. That is, it becomes possible to simplify the operation necessary for the entry registration, it is possible to reliably improve the convenience. In addition, even when the entry/exit state determination unit **110** determines that the state is the post-entry state, the actual intention of the driver may differ from that of the driver, for example, when the driver is scheduled to restart the vehicle VH. In such cases, the display device **61** displays the registration confirmation screen of the other registration mode in response to the touch operation of the switching button B**2** of the driver, so that the presentation of the registration mode suitable for the driver's intention can be realized.

[0038] The registration confirmation processing unit **120** displays, on the display device **61**, the post-exit registration confirmation screen G**2** shown in FIG. **4** when the entry/exit state determination unit **110** determines that the post-exit state is satisfied (that is, when the third specific condition is satisfied). The post-exit registration confirmation screen G**2** includes, for example, an exit path R**2** schematically showing a portion required to exit the most recent travel path stored in RAM **13** or the like by the travel path acquisition unit **100**, an exit registration confirmation button B**3**, and a switching button B**2**. When the driver touches the exit registration confirmation button B**3**, the registration confirmation processing unit **120** transmits a registration command for storing the exit path R**2** in association with the position information of the parking space to the parking space storage processing unit **130**. On the other hand, when the driver touches the switching button B**2**, the registration confirmation processing unit **120** displays the registration confirmation screen G**1**, G**3**A, G**4**A of another registration mode on the display device **61**.

[0039] In this way, when the entry/exit state determination unit **110** determines the post-exit state, the display device **61** displays the post-exit registration confirmation screen G**2** including the exit registration confirmation button B**3** and the exit path R**2**. As a result, the driver can easily perform the exit registration without selecting whether the registration operation is before or after the exit, or whether the travel path is the entry path or the exit path. That is, it is possible to simplify the operation necessary for the exit registration, it is possible to reliably improve the convenience. Even when the entry/exit state determination unit **110** determines that the state is the post-exit state, there is a case where the actual intention of the driver is different from that of the driver, such as a case where the driver is scheduled to reenter. In such cases, the display device **61** displays the registration confirmation screen of the other registration mode in response to the touching operation of the switching button B**2**, so that the presentation of the registration mode suitable for the driver's intention can be realized.

[0040] The registration confirmation processing unit **120** displays, on the display device **61**, the pre-exit registration confirmation screen G**3**A shown in FIG. **5**A when the entry/exit state determination unit **110** determines that the pre-exit state (that is, when the second specific condition or the fifth specific condition is satisfied). The pre-exit registration confirmation screen G**3**A includes, for example, an exit registration starting button B**4** and a switching button B**2**. When the driver touches the exit registration starting button B**4**, the registration confirmation processing unit **120** sends a command to acquire a travel path and store it in the RAM **13** etc. to the travel path acquisition unit **100**. On the other hand, when the driver touches the switching button B**2**, the registration confirmation processing unit **120** displays the registration confirmation screen G**1**, G**2**, G**4**A of another registration mode on the display device **61**.

[0041] As described above, when the entry/exit state determination unit **110** determines that the pre-exit state, a pre-exit registration confirmation screen G**3**A including the exit registration starting button B**4** is displayed on the display device **61**. Thus, the driver can easily start the exit registration without selecting whether the registration operation is before or after exit. That is, it is possible to simplify the operation necessary to start the exit registration, it is possible to reliably improve the convenience. In addition, even when the entry/exit state determination unit **110** determines that the state is the pre-exit state, there is a case in which the actual intention of the driver is different. In such cases, the display device **61** displays the registration confirmation screen of the other registration mode in response to the touching operation of the switching button B**2**, so that the presentation of the registration mode suitable for the driver's intention can be realized.

[0042] When the exit completion condition of the vehicle VH is satisfied, the registration confirmation processing unit **120** displays the exit completion confirmation screen G**3**B shown in FIG. **5**B on the display device **61**. The exit completion confirmation screen G**3**B includes, for example, an exit path R**2** schematically showing a part required to exit the most recent travel path stored in RAM **13** or the like by the travel path acquisition unit **100**, and an exit registration confirmation button B**3**. The exit completion condition may be established, for example, when the travel distance from the start of exit of the vehicle VH reaches a predetermined threshold distance or when the driver touches an exit end button (not shown) displayed on the display device **61** after the vehicle VH starts to exit. When the driver touches the exit registration confirmation button B**3** on the exit completion confirmation screen G**3**B, the registration confirmation processing unit **120** transmits a registration command for storing the exit path R**2** in association with the position information of the parking space to the parking space storage processing unit **130**.

[0043] In this way, the entry/exit state determination unit **110** determines the pre-exit state, and when the exit completion condition is satisfied, the exit completion confirmation screen G**3**B including the exit registration confirmation button B**3** and the exit path R**2** is displayed on the display device **61**. As a result, the driver can easily perform the exit registration without selecting whether the travel path is the entry path or the exit path. That is, it is possible to simplify the operation necessary for the exit registration, it is possible to reliably improve the convenience.

[0044] The registration confirmation processing unit **120** displays the pre-entry registration confirmation screen G**4**A shown in FIG. **6**A on the display device **61** when the entry/exit state determination unit **110** determines that the pre-entry state is satisfied (that is, when the fourth specific condition is satisfied). The pre-entry registration confirmation screen G**4**A includes, for example, an entry registration starting button B**5** and a switching button B**2**. When the driver touches the entry registration starting button B**5**, the registration confirmation processing unit **120** sends a command to acquire the travel route and store it in the RAM **13** to the travel path acquisition unit **100**. On the other hand, when the driver touches the switching button B**2**, the registration confirmation processing unit **120** displays a registration confirmation screen of another registration mode on the display device **61**.

[0045] As described above, when the entry/exit state determination unit **110** determines the pre-entry state, the pre-entry registration confirmation screen G**4**A including the entry registration starting button B**5** is displayed on the display device **61**. As a result, the driver can easily start the entry registration without selecting whether the registration operation is before or after entry. That is, it is possible to simplify the operation necessary for starting the entry registration, it is possible to reliably improve the convenience. Further, even when the entry/exit state determination unit **110** determines that the state is the pre-entry state, there is a case where the actual intention of the driver is different. In such cases, the display device **61** displays the registration confirmation screen of the other registration mode in response to the touching operation of the switching button B**2**, so that the presentation of the registration mode suitable for the driver's intention can be realized.

[0046] When the entry completion condition of the vehicle VH is satisfied, the registration confirmation processing unit **120** displays the entry completion confirmation screen G**4**B shown in FIG. **6**B on the display device **61**. The entry completion confirmation screen G**4**B includes, for example, an entry path R**1** schematically showing a portion required for entry out of the most recent travel path stored in RAM **13** or the like by the travel path acquisition unit **100**, and an entry registration confirmation button B**1**. The entry completion condition may be established, for example, when the shifting position of the vehicle VH is switched to the parking position P. When the driver touches the entry registration confirmation button B**1** on the entry completion confirmation screen G**4**B, the registration confirmation processing unit **120** transmits a registration command for storing the entry path R**1** in association with the position information of the parking space to the parking space storage processing unit **130**.

[0047] As described above, when the entry/exit state determination unit **110** determines the pre-entry state and the entry completion condition is satisfied, the entry completion confirmation screen G**4**B including the entry path R**1** is displayed on the display device **61**. As a result, the driver can easily perform the entry registration without selecting whether the travel path is the entry path or the exit path. That is, it becomes possible to simplify the operation necessary for the entry registration, it is possible to reliably improve the convenience.

[0048] Upon receiving the registration command from the registration confirmation processing unit **120**, the parking space storage processing unit **130** performs a parking space storage process of storing the position information of the parking space as a registered parking space in association with the entry path R**1** and the exit path R**2**, for example, in a storage medium such as a ROM **12**. For example, the position information of the vehicle VH acquired by the position information acquisition device **50** may be used as the position information of the parking space.

[0049] When the distance between the current position of the vehicle VH and the registered parking space becomes equal to or less than the predetermined distance based on the detection result from the position information acquisition device **50**, the parking space notification processing unit **140** performs a notification process of notifying the occupant of the vehicle VH that the registered parking space has been detected. The notification to the occupant may be performed by, for example, a display on the display device **61** or a sound from the speaker **62**.

[0050] When the parking space notification processing unit **140** performs the notification process and the automatic park switch **70** is turned on by the occupant of the vehicle VH (for example, a driver), the automatic park control unit **150** causes the vehicle VH to travel along the target travel path with the entry path R**1** or the exit path R**2** stored in association with the registered parking lot as the target travel path. Accordingly, automatic park control is performed in which the vehicle VH is automatically caused to enter the registered parking space or is automatically caused to exit the registered parking space. Automatic park control is performed by, for example, feedback-controlling the operation of the drive device **20**, the steering device **21**, or the like based on the deviation between the target travel path and the actual travel path of the vehicle VH. The actual moving trajectory of the vehicle VH may be acquired, for example, by odometry based on the detection results of the vehicle speed sensor **31** and the steering angle sensor **34**. Alternatively, the actual moving trajectory of the vehicle VH may be used in combination with the position information of the vehicle VH acquired by the position information acquisition device **50**. Alternatively, the actual moving trajectory of the vehicle VH may be acquired by optical flow based on road surface images or the like of the surroundings of the vehicle VH captured by the camera sensor **43**.

[0051] Next, based on the flow chart shown in FIG. **7**, the routine of the entry/exit state determination process and the registration confirmation process that are performed by the CPU **11** of the ECU **10** will be described. In the following description, for the sake of convenience, a case in which the driver does not touch the switching button B**2** of the registration-confirmation screen displayed on the display device **61** will be described.

[0052] In S**100**, ECU **10** determines whether or not a register-start-request has been received from the drivers. When the register start-request is received (Yes), ECU **10** proceeds to S**110** process. On the other hand, if the register start-request is not accepted (No), ECU **10** returns this routine.

[0053] In S**110**, ECU **10** determines whether the shifting position is the parking position P. If the shifting position is the parking position P (Yes), ECU **10** proceeds to S**120** process. On the other hand, if the shifting position is not the parking position P (No), ECU **10** proceeds to S**160** process.

[0054] When the process proceeds from S**110** to S**120** process, ECU **10** determines whether or not there is a travel history (travel path acquired by the travel path acquisition unit **100**). If there is a travel history (Yes), ECU **10** proceeds to S**130** process. On the other hand, if there is no travel history (No), ECU **10** proceeds to S**150** process.

[0055] When the process proceeds from S**120** to S**150** process, ECU **10** determines that the vehicle VH is in the "pre-exit state" immediately before the vehicle exits the predetermined parking space (second specific condition is satisfied). Next, in S**152**, ECU **10** displays the pre-exit registration confirmation screen G**3**A on the display device **61**. When the driver touches the exit registration starting button B**4**, ECU **10** proceeds to S**154** process and determines whether the exit completion condition is satisfied. When the exit completion condition is not satisfied (No), ECU **10** returns to S**154** process. On the other hand, when the exit completion condition is satisfied (Yes), ECU **10** proceeds to S**156** process.

[0056] In S**156**, ECU **10** displays the exit completion confirmation screen G**3**B on the display device **61**. When the driver touches the exit registration confirmation button B**3**, ECU **10** proceeds to S**158** process, performs a registration process of storing the exit path R**2** in association with the position information of the parking space, and returns this routine.

[0057] When proceeding from S**120** to S**130** process, ECU **10** determines whether or not a predetermined period or more has elapsed from when the shift position is switched to the parking position P until the driver registration start request is received. When the predetermined period or more has elapsed (Yes), that is, when the fifth specific condition is satisfied, ECU **10** proceeds to S**150** process. On the other hand, when the predetermined period or more has not elapsed (No), ECU **10** proceeds to S**140** process.

[0058] When the process proceeds from S**130** to S**140** process, ECU **10** determines that the vehicle VH is in the "post-entry state" immediately after entering the predetermined parking space (first specific condition is satisfied). Next, in S**142**, ECU **10** displays the post-entry registration confirmation screen G**1** on the display device **61**. When the driver touches the entry registration confirmation button B**1**, the ECU **10** proceeds to S**144** process, performs a registration process of storing the entry path R**1** in association with the position information of the parking space, and returns this routine.

[0059] When the process proceeds from S**110** to S**160** process, ECU **10** determines whether or not the travel distance of the vehicle VH after the shift position is switched from the parking position P is less than or equal to a predetermined threshold distance. If the travel distance is less than or equal to the predetermined threshold distance (Yes), ECU **10** proceeds to S**170** process. On the other hand, if the travel distance is not less than the predetermined threshold distance (No), ECU **10** proceeds to S**180** process.

[0060] When the process proceeds from S**160** to S**170** process, ECU **10** determines that the vehicle VH is in the "post-exit state" immediately after the vehicle exits the predetermined parking space (third specific condition is satisfied). Next, in S**172**, ECU **10** displays the post-exit registration confirmation screen G**2** on the display device **61**. When the driver touches the exit registration confirmation button B**3**, ECU **10** proceeds to S**174** process, performs a registration process of storing the exit path R**2** in association with the position information of the parking space, and returns this routine.

[0061] When the process proceeds from S**160** to S**180** process, ECU **10** determines that the vehicle VH is in the "pre-entry condition" immediately before the vehicle enters the predetermined parking space (fourth specific condition is satisfied). Next, in S**182**, ECU **10** displays the pre-entry registration confirmation screen G**4**A on the display device **61**. When the driver touches the entry registration starting button B**5**, ECU **10** proceeds to S**184** process and determines whether the entry completion condition is satisfied. When the entry completion condition is not satisfied (No), ECU **10** returns to S**184** process. On the other hand, when the entry completion condition is satisfied (Yes), ECU **10** proceeds to S**186** process.

[0062] In S**186**, ECU **10** displays the entry completion confirmation screen G**4**B on the display device **61**. When the driver touches the entry registration confirmation button B**1**, ECU **10** proceeds to S**188** process, performs a registration process of storing the entry path R**1** in association with the position information of the parking space, and returns this routine.

[0063] It should be noted that the present disclosure is not limited to the above-described embodiments, and various modifications can be made without departing from the scope of the present disclosure.

[0064] For example, in the above embodiment, the driver performs the registration start request, but it is also possible to configure the system side to perform the registration start request. For example, when the position information of the parking space that the vehicle VH is caused to enter or exit, and the vehicle VH is caused to enter the parking space that is frequently used, or when the vehicle VH is caused to exit the parking space, a registration start request is made on the system side without requiring the driver's operations, and the registration confirmation screen described above is displayed on the display device **61**. With this configuration, it is possible to further reduce the operation load of the driver related to registration. The present disclosure can also be applied to an autonomous vehicle that automatically performs some or all of the driving operations.

What is claimed is:

1. A park assist system that is configured to store, as a registered travel path, a travel path of a vehicle when causing the vehicle to enter a predetermined parking space or a travel path of the vehicle when causing the vehicle to exit the parking space, and perform automatic park control when causing the vehicle to enter or exit the parking space from next time onwards, the automatic park control being control in which the vehicle is caused to travel autonomously based on the registered travel path, the park assist system comprising a determination unit configured to

determine whether to store the registered travel path as an entry path or an exit path, based on a state of the vehicle when a registration request to store the registered travel path is made, and

determine whether to store a travel path of the vehicle immediately before or a travel path of the vehicle immediately after when the registration request is made.

2. The park assist system according to claim **1**, further comprising a display processing unit configured to perform a display process of displaying a confirmation screen of a registration mode according to a determination result from the determination unit on a display device installed in a driver's cabin of the vehicle.

3. The park assist system according to claim **1**, wherein the determination unit is configured to, in a case where, when the registration request is made, a shift position of the vehicle is a parking position and there is the travel path immediately before, determine that the registered travel path is the entry path and determine to store the travel path immediately before.

4. The park assist system according to claim **1**, wherein the determination unit is configured to, in a case where, when the registration request is made, a shift position of the vehicle is a parking position and there is not the travel path immediately before, determine that the registered travel path is the exit path and determine to store the travel path immediately after.

5. The park assist system according to claim **1**, wherein the determination unit is configured to, in a case where, when the registration request is made, a shift position of the vehicle is a position other than a parking position and a distance traveled by the vehicle after switching of the shift position from the parking position is a predetermined threshold distance or less, determine that the registered travel path is the exit path and determine to store the travel path immediately before.

6. The park assist system according to claim **1**, wherein the determination unit is configured to, in a case where, when the registration request is made, a shift position of the vehicle is a position other than a parking position and a distance traveled by the vehicle after switching of the shift position from the parking position is greater than a predetermined threshold distance, determine that the registered travel path is the entry path and determine to store the travel path immediately after.

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